

Full SmolVLA: Foundation Action Experts on the SmolVLM-256M Backbone

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Abstract

Full SmolVLA represents the final architectural paradigm in our repository: the direct integration of an end-to-end Vision-Language Model (VLM) foundation backbone (SmolVLM-256M-Instruct). Instead of segregated dual-stream fusion, multimodal reasoning occurs innately within the 30-layer autoregressive transformer backbone. We introduce a novel multi-layer tapping architecture that extracts hierarchical spatial and semantic contexts, routing them into an OT-CFM Action Expert. This framework yields unparalleled generalization capabilities while demanding less than 1 million trainable parameters.

1. Deep Architecture Formulation

The foundation backbone is completely frozen and loaded in hardware-efficient bfloat16 precision. Rather than extracting exclusively from the terminal layer, Full SmolVLA taps intermediate hidden states across layers 10, 20, and 30. Layer 10 provides low-level spatial geometry, Layer 20 provides relational semantic maps, and Layer 30 resolves the global task abstraction.

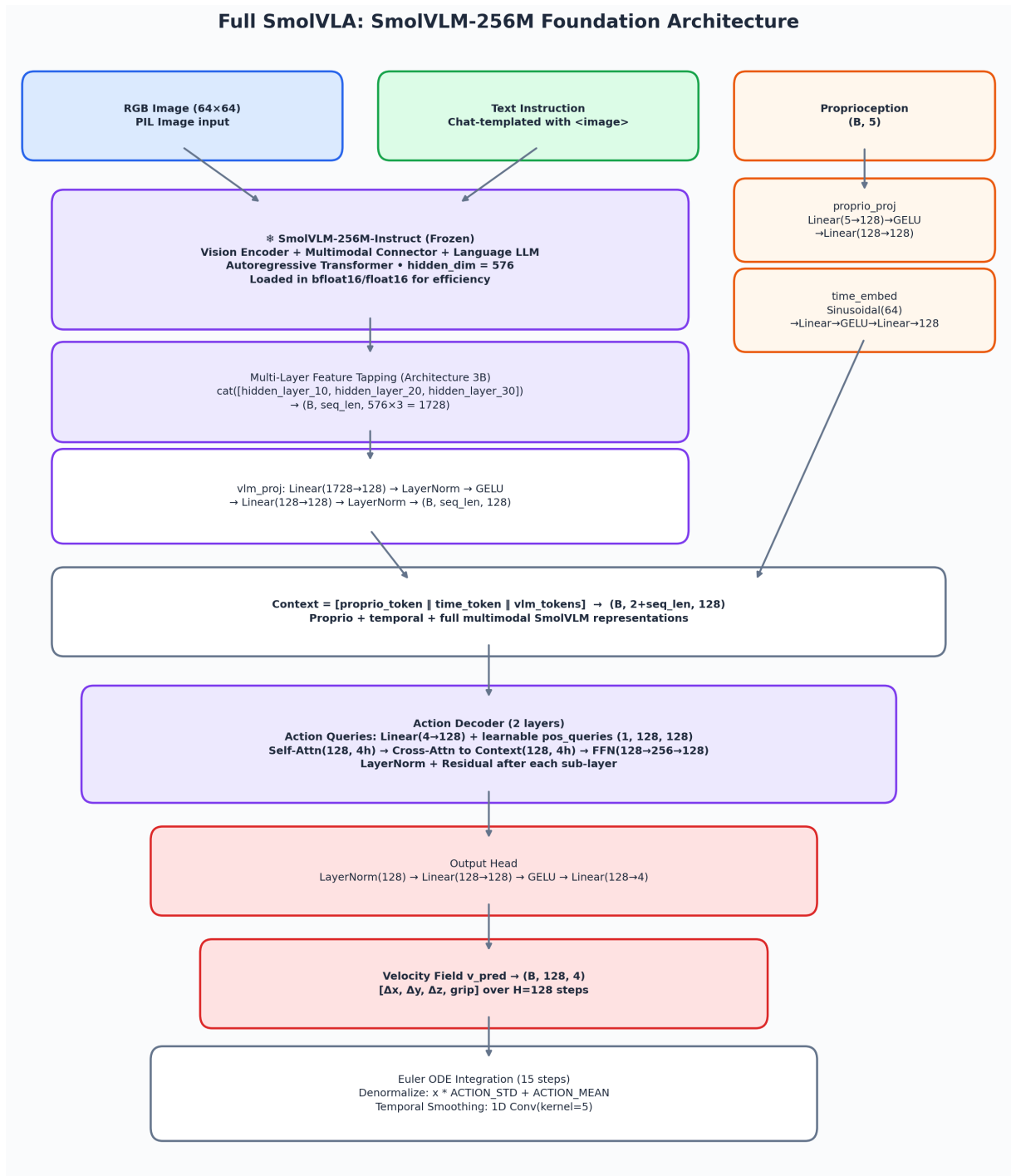


Figure 1: Full SmolVLA Architecture featuring the SmolVLM-256M backbone and intermediate layer tapping.

1.1 Network Graph & Parameter Allocation

The following architectural readout maps the flow from the concatenated 1728-dimensional multi-layer VLM features through the downstream projection network and onto the multi-head action decoder.

Layer (type:depth-idx)	Output Shape	Param #
FullSmolVLAPolicy	[1, 128, 4]	254,720
+Sequential: 1-1	[1, 128]	--
+-Linear: 2-1	[1, 128]	768
+-GELU: 2-2	[1, 128]	--
+-Linear: 2-3	[1, 128]	16,512
+Sequential: 1-2	[1, 128]	--
+-ContinuousSinusoidalTimeEmbedding: 2-4	[1, 64]	--
+-Linear: 2-5	[1, 128]	8,320

+-GELU: 2-6	[1, 128]	--
+-Linear: 2-7	[1, 128]	16,512
+-Linear: 1-3	[1, 128, 128]	640
+-MultiheadAttention: 1-4	[1, 128, 128]	66,048
+-LayerNorm: 1-5	[1, 128, 128]	256
+-MultiheadAttention: 1-6	[1, 128, 128]	66,048
+-LayerNorm: 1-7	[1, 128, 128]	256
+-Sequential: 1-8	[1, 128, 128]	--
+-Linear: 2-8	[1, 128, 256]	33,024
[... intermediate layers omitted ...]		
+-Linear: 2-11	[1, 128, 256]	33,024
+-GELU: 2-12	[1, 128, 256]	--
+-Linear: 2-13	[1, 128, 128]	32,896
+-LayerNorm: 1-15	[1, 128, 128]	256
+-Sequential: 1-16	[1, 128, 4]	--
+-LayerNorm: 2-14	[1, 128, 128]	256
+-Linear: 2-15	[1, 128, 128]	16,512
+-GELU: 2-16	[1, 128, 128]	--
+-Linear: 2-17	[1, 128, 4]	516

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Total params: 712,324
Trainable params: 712,324
Non-trainable params: 0
Total mult-adds (Units.MEGABYTES): 0.19
=====
Input size (MB): 0.02
Forward/backward pass size (MB): 1.97
Params size (MB): 0.77
Estimated Total Size (MB): 2.77
=====

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2. Trajectory Generation and Cross-Attention Mechanics

The Flow Matching Action Expert constructs its trajectory over a horizon of 128 continuous steps. The Action Decoder queries the foundation multi-layer tokens via 128 self-attending temporal query vectors. By inspecting the extracted attention matrices, we can observe precisely which multimodal tokens the action trajectory references during different temporal phases of the robotic motion.

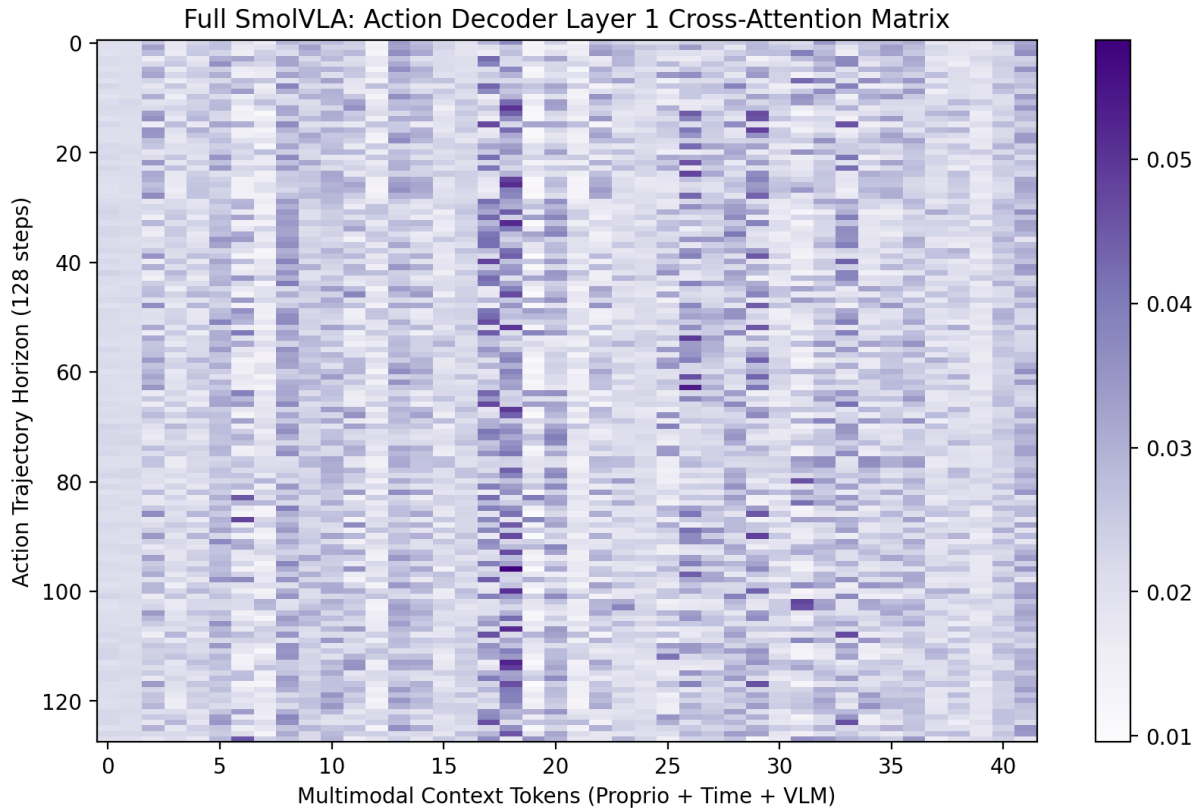


Figure 2: Real Action Decoder Cross-Attention Weights (128 Trajectory Queries vs. VLM Context).

Figure 2 illustrates a live forward-pass extraction of the cross-attention layer. The network dynamically distributes attention: early trajectory states heavily attend to object grounding tokens, while terminal states strongly correlate with destination context and proprioceptive limits.

3. Conditional Flow Optimization

The network synthesizes the temporal ODE vector field using Euler integration. Crucially, by offloading the heavy multimodal inference to a pre-cached offline worker script, the training of the Action Expert completes in a fraction of the time required for traditional end-to-end architectures, rendering Full SmoVLA both deeply comprehensive and computationally lightweight.